



12. What happens when a new instruction needs to be added or an existing one modified in a hardwired control unit?
 A) The CPU requires a complete redesign. B) The control unit's microcode is updated.
 C) A new hardware module is added. D) The control table is modified.
13. The method for updating the main memory as soon as a word is removed from the Cache is called.
 A) Write-through B) write-back
 C) protected write D) intense heat radiations
14. In a three BUS architecture, how many input and output ports are there in register file?
 A) 2 output and 2 input B) 1 output and 2 input
 C) 2 output and 1 input D) 1 output and 1 input
15. A set of microinstructions for a single machine instruction is called _____
 A) Program B) Command
 C) Micro program D) Micro Command
16. When a pipelining includes four phases like 50, 70, 90 & 100 ns duration with 10 ns latch delay, then calculate Non-Pipeline Execution Time?
 A) 250 B) 300
 C) 310 D) 320
17. A special unit used to govern the out of order execution of the instructions is called as _____
 A) Commitment unit B) Temporal unit
 C) Monitor D) Supervisory unit
18. The commitment unit uses a queue called _____
 A) Record buffer B) Commitment buffer
 C) Storage buffer D) None
19. The _____ plays a very vital role in case of super scalar processors.
 A) Compilers B) Motherboard
 C) Memory D) Peripherals
20. What is the maximum number of threads that can be launched in a single CUDA block?
 A) 1024 B) 4096
 C) 20248 D) 8192

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain any 5 addressing modes available in a machine with proper examples.	10	L*2	1	1
	b) Implement 2 address assembly code for the expression $A = A + B * C / E - 4$.	4	L2	1	1
	c) What is the purpose of MAR, MDR registers?	2	L1	2	1
2.	a) Explain how multiplication can be carried out with less number of additions using Booth's algorithm? Multiply -13 x 11 using the same.	7	L2	2	1
	b) Register R1 and R2 of computer contains decimal values 1200 and 4600. What is the effective address of the memory operand in each of the following instructions? Load 20(R1), R5 Move #300, R5 Store R5, 30(R1, R2) Add -(R2), R5 Sub (R1)+, R5.	4	L2	1	1
	c) Explain how floating point numbers are represented using IEEE single precision representation? Represent +3.52 using the same showing the steps involved.	5	L3	2	1
3.	a) With a circuit diagram explain how non restoring binary division can be carried out? Perform Non Restoring binary division for 14 / 3.	10	L2	2	1

- b) Describe the principle of carry look – a head addition for 4 – bit adder circuit using B – cells and calculate the number of gate delays for S3 & C4

6 L2 2 1

Unit – II

4. a) Explain the working of single bus organization of datapath. Show the signal sequence for the execution of Move (R3), R1.
b) Why instructional hazards are important to be solved? Discuss different types of the instructional hazards.

10 L2 3 1

6 L2 4 1

5. a) Explain the working of three bus organization of data path. Show the signal sequence for the execution of Add R4, R5, R6.
b) Write short notes on i) associative caches ii) on-chip parallelism.

10 L2 3 1

6 L2 4 1

6. a) What is the use of cache memory? With diagram explain various methods of cache mapping.
b) How cache coherence is managed in a NUMS multiprocessor. Illustrate with example.

10 L2 4 1

6 L2 4 1

Unit – III

7. a) State AMDAHL's law. Explain different I/O control methods.
b) Explain the role of a CUDA kernel and how it is executed on the GPU. How are threads organized, and what are blocks and grids in this context?
c) Discuss the significance of thread indexing using threadIdx and blockIdx, and how these indices are utilized to manage parallelism in CUDA programs.
8. a) With diagram explain Direct Memory Access (DMA). Demonstrate DMA and its implementation and show how the data is transferred between memory and I/O devices using DMA controller.
b) Explain the concept of data parallelism with an example.
c) What unique architectural elements define a GPU, and how do they enable the GPU's specialization in parallel processing.

6 L1 5 1

5 L2 5 1

5 L2 5 1

6 L2 5 1

4 L1 5 1

6 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
